

Assignment 2: Factor Models, PCA, and Fama–French

Mathematical and Empirical Finance, 2025–2026

Instructions

- You should submit your answers (written in English) as a PDF file in combination with a code file (or a R-Markdown file containing both answers and code), via the Assignment upload tool in Canvas. It should be possible to understand your answers without opening the code file, but you may refer to the details of particular calculations in the code.
- Some questions may require an answer to an earlier question. If you have been unable to fully answer the earlier question, replace this answer by a (reasonable) hypothetical value, and clearly indicate that you will use this guess in the rest of the assignment.
- Indicate the group number on the first page of the report together with the names and student numbers of your group members (maximum three) that were actively present and working in class on the assignment.
- Indicate your group number in the names of all files before submission: use the format “MEFAssignment2GroupN”, where N is your group number between 1 and 52.
- If (and only if) you have problems submitting your work via Canvas, you may e-mail it to m.ferrari@uva.nl, with “MEF Assignment 2, Group N” in the subject line, where N as above.

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Dataset

- The dataset for this assignment is in the `csv`-file `DataAssignment2.csv`, where you can find the daily returns of 22 stocks (columns 2 through 23). The first six stock return series (columns 2 through 7) correspond to the big-tech reference group referred to below, namely `AAPL`, `MSFT`, `AMZN`, `GOOGL`, `META`, and `NVDA`. Moreover, you will find the excess market return (column 24) and the risk-free rate (column 25). In columns 26 and 27, you will find the SMB and HML Fama–French factors, respectively.
- The data were downloaded from Yahoo Finance (stock data) and Ken French’s data library (risk-free rate, market excess return and SMB, HML factors).
- The data range from 2020-01-02 up to 2024-12-31, resulting in 1258 daily observations.
- All rates of return are in percentages. For example, if an entry in the data file reads 2.5, then this means a rate of return of 2.5%.

Factor Models

In this assignment, we are going to analyze stock-return factor models using principal components and the Fama–French 3-factor model.

1. *[20 points]* First, you consider a factor model based on principal component analysis. Using the 22 stock return series, determine how many factors are needed to explain about 90% of the total variation in the return series. If 90% is not matched exactly, choose the number of factors whose cumulative explained variance is closest to 90%, even if it is below 90%. Also provide the scree plot.
2. *[35 points]* Having determined the number of principal components, compare a factor model based on these principal components with the Fama–French 3-factor model. Run the regressions for *all* 22 stocks. Then, focusing on the big-tech reference group only, compare the two factor models in terms of both average adjusted regression R^2 and residual correlation. In particular, compute for each model the average adjusted R^2 across the first six stocks, as well as the average absolute off-diagonal residual correlation for these six stocks. Briefly comment on what these two measures suggest about the fit of the two models for the big-tech reference group.

3. *[25 points]* Compute the covariance matrices of stock returns implied by the two factor models under consideration. For each model, report the 6×6 covariance submatrix corresponding to the first six stocks, i.e., to the big-tech reference group. For comparison, also report the corresponding 6×6 sample covariance submatrix. To quantify the difference, compute for each model the Frobenius distance between its implied covariance submatrix and the sample covariance submatrix. Based on this measure, determine which factor model provides the better covariance fit for the reference group, and briefly justify your conclusion.
4. *[20 points]* Construct the first three principal components and compare them quantitatively to the three Fama–French factors: the excess market return R_M^e , SMB, and HML. In particular, report the 3×3 matrix containing the correlations between $(PC1, PC2, PC3)$ and $(R_M^e, \text{SMB}, \text{HML})$. Which of the first three principal components are correlated with which Fama–French factors? Also discuss whether some Fama–French factors are not well captured by the first three principal components.